

## Problem Set 8 (50 pts)

Due at 11:59pm

Friday, October 31

All the problems are from <https://aimacode.github.io/aima-exercises/> (possibly) with minor typo fixes, slight rewriting, or additions of missing information.

**1.(8 pts)** (Exercise 9.4) For each pair of atomic sentences, give the most general unifier if it exists:

(a) **(2 pts)**  $P(A, B, B), P(x, y, z)$ .

(b) **(2 pts)**  $Q(y, G(A, B)), Q(G(x, x), y)$ .

(c) **(2 pts)**  $Older(Father(y), y), Older(Father(x), John)$ .

(d) **(2 pts)**  $Knows(Father(y), y), Knows(x, x)$ .

**2.(18 pts)** (Exercise 9.23) Suppose a knowledge base contains just the following first-order Horn clauses:

$$\begin{aligned} &Ancestor(Mother(x), x) \\ &Ancestor(x, y) \wedge Ancestor(y, z) \Rightarrow Ancestor(x, z) \end{aligned}$$

Consider a forward chaining algorithm that, on the  $j$ th iteration, terminates if the KB contains a sentence that unifies with the query, else adds to the KB every atomic sentence that can be inferred from the sentences already in the KB after iteration  $j - 1$ .

(a) **(12 pts)** For each of the following queries, say whether the algorithm will (1) give an answer (if so, write down that answer); or (2) terminate with no answer; or (3) never terminate.

(i)  $Ancestor(Mother(y), John)$

(ii)  $Ancestor(Mother(Mother(y)), John)$

(iii)  $Ancestor(Mother(Mother(Mother(y))), Mother(y))$

(iv)  $Ancestor(Mother(John), Mother(Mother(John)))$

**(b)(6 pts)** Can a resolution algorithm prove the sentence  $\neg \text{Ancestor}(\text{John}, \text{John})$  from the original knowledge base? Explain how, or why not.

**3.(24 pts)** (Exercise 9.29) Here are two sentences in the language of first-order logic:

$$(A) \quad \forall x \exists y \, x \geq y$$

$$(B) \quad \exists y \forall x \, x \geq y$$

Assume that the variables range over all the natural numbers  $0, 1, 2, \dots$ , and that the “ $\geq$ ” predicate means “is greater than or equal to.”

**(a) (3 pts)** Translate (A) and (B) into English.

**(b) (2 pts)** Is (A) true under this interpretation?

**(c) (2 pts)** Is (B) true under this interpretation?

**(d) (3 pts)** Does (A) logically entail (B)?

**(e) (3 pts)** Does (B) logically entail (A)?

**(f) (5 pts)** Using resolution, try to prove that (A) follows from (B). Do this even if you think that (B) does not logically entail (A); continue until the proof breaks down and you cannot proceed (if it does break down). Show the unifying substitution for each resolution step. If the proof fails, explain exactly where, how, and why it breaks down.

**(g) (6 pts)** Now try to prove that (B) follows from (A).